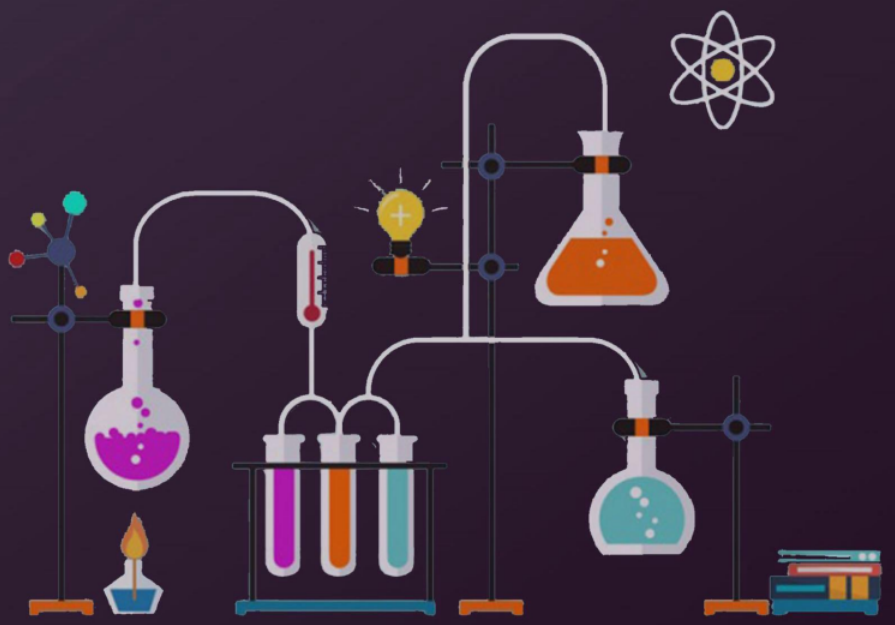
CHEMISTRY

p-Block Group (15-18)



SHEET

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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr.No.	Topic Name	Ex-1	Ex-1A	Ex-2	Ex-2A
1	Occurence and Physical Properties of Nitrogen Family	1	13	1,2	
2	Chemical Properties Of Group 15 Elements			3	
3	Dinitrogen				
4	Ammonia	2			
5	Oxides Of Nitrogen And Nitrous Acid	3,4	1,2,7-10	4	1
6	Nitric Acid	5	3		
7	Allotropic Forms of Phosphorous				
8	Important Compounds of Phosphorus	6,7	11	5-7	2,3, 15-17,36
9	Introduction To Group 16 Elements		4	8	
10	Chemical Properties of Group 16 Elements	8		9	4-6
11	Dioxygen				
12	Simple Oxides and Ozone	9	14		7
13	Allotropes of Sulphur and Sulphur Dioxide			10-11	
14	Hydrogen Sulphide and Sulphuric Acid			12-14	
15	Introduction to Group 17 Elements	10-16	18,19		
16	Chemical Properties of Halogen Family	17-21	12,15	15-16	33
17	Fluorine and Chlorine	22-24	5	17-18	8,18-21, 34,37
18	Hydrochloric Acid				
19	Interhalogen Compounds		16		25-27
20	Introduction to Group 18 Elements		17		
21	Chemical Properties and Uses of Group 18		6	19	9,10
22	Oxyacids and Their Naming			20,21	11,12
23	ACC	25,26		22-25	13,14,22-24, 28-32, 35

Exercise - 1 Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Occurence and Physical Properties of Nitrogen Family

Q1. The correct order of decreasing Lewis acid strength of trichlorides of group 15 elements is -

(A) $AsCl_3 > PCl_3 > BiCl_3 > SbCl_3$

(B) $PCl_3 > AsCl_3 > SbCl_3 > BiCl_3$

(C) $NCl_3 > PCl_3 > AsCl_3 > SbCl_3$

(D) $SbCl_3 > BiCl_3 > PCl_3 > AsCl_3$

Ammonia

Q2. Moist ammonia gas is dried over –

(A) P_2O_5

(B) Anhydrous $CaCl_2$

(C) Quick lime

(D) Conc. H_2SO_4

Oxides of Nitrogen and Nitrous Acid

Q3. The hybridisation of atomic orbitals of nitrogen in NO_2^+, NO_3^- and NH_4^+ are –

(A) sp, sp^3 and sp^2 respectively

(B) sp, sp^2 and sp^3 respectively

(C) sp^2, sp and sp^3 respectively

(D) sp^2, sp^3 and sp respectively

Q4. The correct order for the increasing acidic strength of oxides of nitrogen is –

(A) $N_2O < NO < N_2O_4 < N_2O_3 < N_2O_5$

(B) $N_2O < NO < N_2O_3 < N_2O_4 < N_2O_5$

(C) $N_2O < N_2O_3 < N_2O_4 < N_2O_5 < NO$

(D) $NO < N_2O < N_2O_3 < N_2O_4 < N_2O_5$

Q5. $NH_4NO_3 \xrightarrow{heat} N_2O + 2H_2O$

In the reaction shown above-

(A) The oxidation number of nitrogen is changed from –3 to +3.

(B) NH_3 is another side product.

(C) The anion is more oxidising and NH_4^+ ion is oxidised to nitrogen in the +1 state.

(D) The compound decomposes in the way shown, when used as a fertilizer.

Important Compound of Phosphorus

Q6. In the compounds of the type POX_3, the P atom multiple bond is of the type –

(A) $p\pi - p\pi$ (B) $d\pi - d\pi$

(C) $p\pi - d\pi$ (D) None of these

Q7. Which among the following is least acidic–

(A) P_4O_6 (B) P_4O_{10}

(C) As_4O_6 (D) As_4O_{10}

Chemical Properties of group 16 Elements

Q8. Which of the following hydrides is most acidic –

(A) H_2Te (B) H_2Se (C) H_2O (D) H_2S

Simple Oxides and Ozone

Q9. Sequence of acidic character is –

(A) $SO_2 > CO_2 > CO > N_2O_5$

(B) $SO_2 > N_2O_5 > CO > CO_2$

(C) $N_2O_5 > SO_2 > CO > CO_2$

(D) $N_2O_5 > SO_2 > CO_2 > CO$

Introduction to group 17 Elements

Q10. Which of the following polyhalide ion is most stable –

(A) Br_3^- (B) I_3^- (C) Cl_3^- (D) F_3^-

Q11. The bonds between H and X have a degree of polarity that follows the order –

(A) H – F > H – Cl > H – Br > H – I

(B) H – F > H – Cl > H – I > H – Br

(C) H – F > H – I > H – Cl > H – Br

(D) H – I >H – F > H – Cl > H – Br

Q12. Which of the following is most basic –

(A) F^- (B) Cl^-

(C) Br^- (D) I^-

Q13. The ionic character of H – X is highest in –

(A) HBr (B) HI

(C) HF (D) HCl

Q14. The strongest acid is –

(A) H_3PO_2

(B) H_3PO_3

(C) H_3PO_4

(D) All are strongest acid

Q15. Strength of halogen acids, HF, HCl, HBr and HI varries in the order –

(A) HF > HCl > HBr > HI

(B) HF > HBr > HI > HCl

(C) HBr > HI > HCl > HF

(D) HI > HBr > HCl > HF

Q16. Which of the following HY (hydra-acid of halogens) is the strongest acid, best reducing agent, is the least stable, and is easly oxidised –

(A) HF (B) HCl

(C) HBr (D) HI

Chemical Properties of Halogen Family

Q17. Conjugate of which of the following a pyramidal.

(A) $ClO_3(OH)$ (B) $ClO_2(OH)$

(C) ClO(OH) (D) Cl(OH)

Q18. F_2 reacts with dilute NaOH to give x formula of x will be?

(A) F_2O (B) OF_2

(C) F_2O_2 (D) F_2O_3

Q19. The order of stability of oxides of chlorine is -

(A) $Cl_2O_6 < ClO_2 < Cl_2O < Cl_2O_7$

(B) $Cl_2O > ClO_2 > Cl_2O_6 > Cl_2O_7$

(C) $Cl_2O_7 > Cl_2O_6 > ClO_2 > Cl_2O$

(D) $Cl_2O_7 > ClO_2 > Cl_2O > Cl_2O_6$

Q20. Which is correct order of basic strength –

(A) $OCl^- > ClO_2^- > ClO_3^- > ClO_4^-$

(B) $ClO_4^- > ClO_3^- > ClO_2^- > ClO^-$

(C) $ClO_3^- > ClO_4^- > ClO_2^- > ClO^-$

(D) None of these

Q21. Strong oxidising agent and strong reducing agents are respectively –

(A) F_2, HI (B) Cl_2, HCl

(C) Br_2, HBr (D) I_2, HF

Fluorine and Chlorine

Q22. The type(s) of bond(s) present in the products formed when excess of chlorine reacts with ammonia.

(A) covalent, dative, ionic

(B) covalent

(C) covalent & dative

(D) ionic

Q23. The shape of ClO_3^- is -

(A) Tetrahedral (B) Triangular planar

(C) Triangular (D) Planar

Q24. Fluorine does not form oxyacids because –

(A) It is most electronegative

(B) It is the most powerful oxidising agent

(C) It lacks d-orbitals

(D) None of these

Q25. XeF_6 on complete hydrolysis gives–

(A) $XeOF_4$ (B) XeO_2F_2

(C) XeO_3 (D) XeO_4

Q26. The ease of liquefaction of noble gases decrease in the order–

(A) He > Ne > Ar > Kr > Xe

(B) He < Ne < Ar < Kr < Xe

(C) He < Ne < Ar > Kr > Xe

(D) He > Ne > Ar < Kr < Xe

Exercise - 1A *Concept Builder*

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Which of the following order is incorrect,

(A) $NH_3 < PH_3 < ASH_3 < SbH_3$ = Reducing nature

(B) $NH_3 < PH_3 < ASH_3 < SbH_3$ = Bond angle

(C) $NH_3 < PH_3 < ASH_3 < SbH_3$ = Basic strength

(D) $NH_3 < PH_3 < ASH_3 < SbH_3$ = Boiling point

Q2. Which type of reactions given nitrous oxides gas.

(A) $4NH_3 + 5O_2 \xrightarrow{Pt\Delta}$

(B) $3Cu + 8HNO_3\,(dil.) \longrightarrow$

(C) $2Pb(NO_3)_2 \xrightarrow{\Delta}$

(D) $Cu + 4HNO_3\,(conc.) \longrightarrow$

Q3. Which of the following statements is/are correct –

(A) OF_2, Cl_2O and Br_2O are bent with two positions occupied by two lone pairs of electrons.

(B) In OF_2, the bond angle is less than 109°28'.

(C) In Cl_2O, the bond angle is greater than 109°28'.

(D) In Br_2O the bond angle is greater than 111°.

Q4. Which of the following statements is/are correct –

(A) Chalcogens are ore forming elements.

(B) When the chalcogen atom is in the least electronegative atom in the molecule, it shows as oxidation state of +2.

(C) Oxygen is never more than divalent.

(D) None is correct.

Q5. Some oxyacids of nitrogen are given, find out monobasic acids :–

(A) Hyponitrous acid (B) Nitrous acid

(C) Nitric acid (D) Pernitric acid

Q6. Choose the correct statements about the compound XeF_4 –

(A) It was the first reported stable binary compound of a noble gas

(B) It is a Lewis base

(C) It has four unshared valence electrons

(D) It has twelve valence electrons in total

PARAGRAPH TYPE

This section contains ***PARAGRAPHS****. Based on each paragraph, there are questions. Each question has FOUR options.* ***ONLY ONE*** *of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph For. Q.7 to 10

An inorganic salt (A) is decomposed at about 523 K to give products (B) and (C). Compound (C) is a liquid at room temperature and is neutral to litmus paper while oxide (B) on burning with white phosphorous, given a dehydrating agent (D).

Q7. Compound (A) is -

(A) NH_4NO_3 (B) N_2O

(C) H_2O (D) P_2O_5

Q8. Compound (B) is -

(A) NH_4NO_3 (B) N_2O

(C) H_2O (D) P_2O_5

Q9. Compound (C) is -

(A) NH_4NO_3 (B) N_2O

(C) H_2O (D) P_2O_5

Q10. Compound (D) is -

(A) NH_4NO_3 (B) N_2O

(C) H_2O (D) P_2O_5

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q11. Match the following

Column I (Reactants)	**Column II (Products)**
(A) Cu + dil HNO_3	(P) NO
(B) Cu + conc HNO_3	(Q) NO_2
(C) Zn + dil HNO_3	(R) N_2O
(D) Zn + conc HNO_3	(S) $Cu(NO_3)_2$ $Zu(NO_3)_2$

Q12. Match the column –

Column I	**Column II**
(A) Cl_2O	(P) Exists in equilibrium with monomer
(B) ClO_2	(Q) Paramagnetic
(C) Cl_2O_6	(R) Strongest acidic oxide
(D) Cl_2O_7	(S) Maximum steric increases bond angle crowding

ASSERTION AND REASON

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

Q13. **Statement 1 :** Nitrogen show catenation properties more than phosphorus.

Statement 2 : Nitrogen shows poor catenation tendency as compared to phosphorus due to lower N – N bond enthalpy.

Q14. **Statement 1 :** Among hydrides of the elements of group 16, water shows unusual physical properties.

Statement 2 : Water shows unusual physical properties due to hydrogen bonding between the water molecules.

Q15. **Statement 1 :** Fluorine is a stronger oxidising agent than chlorine.

Statement 2 : Fluorine can displace chlorine from chlorides.

Q16. **Statement 1 :** Liquid IF_5 conducts electricity.

Statement 2 : Liquid IF_5 self ionizes as, $2\ IF_5 \rightleftharpoons IF_4^+ + IF_6^-$.

Q17. **Statement 1 :** Most of the known noble gas compounds are those of xenon.

Statement 2 : The ionisation enthalpy of xenon is very close to that of oxygen.

SUBJECTIVE TYPE

Q18. Arrange the following in the order or property indicated for each set :

(a) F_2, Cl_2, Br_2, I_2 - increasing bond dissociation enthalpy.

(b) HF, HCl, HBr, HI -increasing acid strength.

(c) NH_3, PH_3, AsH_3, SbH_3, BiH_3 - increasing base strength.

Q19. Arrange the following as indicated.

(i) P_2O_5, N_2O_5, As_2O_5, Bi_2O_5, Sb_2O_5 (decreasing acid strength)

(ii) PF_3, PCl_3, PBr_3, PI_3 (decreasing Lewis acid strength)

(iii) H_3PO_3, H_3PO_4, H_3PO_2 (increasing order of acid strength)

(iv) NCl_3, NF_3, NBr_3, NI_3 (decreasing order of Lewis base)

(v) NH_3, PH_3, AsH_3, SbH_3 (increasing order of HMH bond angle)

Exercise - 2 *Concept Builder*

SINGLE CORRECT TYPE

Each question has FOUR options for correct answer. ***ONLY ONE*** *of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. In the nitrogen family the H – M – H bond angle in the hydrides MH_3 gradually becomes closer to 90° on going from N to Sb. This shows that gradually -

(A) The basic strength of hydrides increases.

(B) Almost pure p-orbitals are used for M – H bonding.

(C) The bond energies of M – H bond increases.

(D) The bond pairs of electrons becomes nearer to the central atom.

Q2. Dipole moment is highest for -

(A) NCl_3 (B) NBr_3 (C) NI_3 (D) NF_3

Q3. As we proceed from NH_3 to BiH_3, which of the following statements is not correct-

(A) The hydrides become increasingly difficult to prepare.

(B) The reducing power of the hydrides increases.

(C) Their stability decreases.

(D) The ease of replacing hydrogen of the hydrides by other group increases.

Q4. $NH_4ClO_4 + HNO_3$ (dilute) $\rightarrow X + HClO_4$

$X \xrightarrow{\text{heat}} Y$ (gas)

Gas (Y) is –

(A) O_2 (B) N_2

(C) NO_2 (D) N_2O

Q.5 In which of following option the (X – A – X) angle difference in cationic part and anionic part is maximum in the solid state.

(A) PCl_5 (B) Cl_2O_6

(C) N_2O_5 (D) BeH_2

Q6. Which of the following statements are not true for phosphorus –

(A) It is non-metal.

(B) It exists as a tetrahedral molecular solid.

(C) It is less reactive than nitrogen.

(D) P – P bond is much weaker than N ≡ N bond.

Q7. The number of P – O – P linkage present in P_4O_{10}, $P_4O_{12}^{4-}$ and $P_4O_{13}^{6-}$ respectively are –

(A) 4, 4, 3 (B) 4, 6, 3

(C) 3, 5, 4 (D) 6, 4, 3

Q8. The molecule having the smallest bond angle is –

(A) H_2O (B) H_2S

(C) H_2Te (D) H_2Se

Q9. A yellow metallic powder is burnt in a stream of fluorine to obtain a colourless gas X which is thermally stable and chemically inert. Its molecule has octahedral geometry. Another colourless gas Y with same constituent atoms as that of X is obtained when sulphur dichloride is heated with sodium fluoride. Its molecule has see-saw structure X and Y are –

(A) SF_4 and S_2F_2 (B) SF_6 and SF_4

(C) NaF and NaCl (D) SF_4 and SF_6

Q10. A gas 'X' bleaches the colour of flowers by reduction and gas 'Y' by oxidation. The X and Y may be :

(A) SO_2 and Cl_2 (B) CO and Cl_2

(C) NH_3 and SO_2 (D) H_2S and Br_2

Q11. A diamagnetic element burns in air to form a dioxide. This dioxide can reduce iodine to HI. When H_2 gas is bubbled through this element, it forms a hydride. It was found that the reaction of the dioxide and hydride of this element produces the same metal again. The element is –

(A) S (B) Si

(C) Te (D) Po

Q12. Which of the following parent oxy acid does not have its pyro–oxy acid.

(A) H_2SO_3 (B) HNO_3

(C) H_3PO_3 (D) H_4SiO_4

Q13. Which of the following is incorrect about H_2SO_4 –

(A) It can act as oxidising agent with copper.

(B) It acts as dehydrating agent with sugar.

(C) It acts as acid in its reaction with BaO_2.

(D) It acts as reducing agent in its reaction with Cu_2O.

Q14. Which of the following oxide is most acidic–

(A) Na_2O (B) Al_2O_3

(C) P_2O_5 (D) SO_3

Q15. Oxygen does not form OF_6 because –

(A) It has a small size

(B) There are no vacant d-orbitals available

(C) It has high ionisation energy

(D) It has large size

Q16. Which reaction can not be used for the preparation of halogen acid –

(A) $2KBr + H_2SO_4\,(conc.) \rightarrow K_2SO_4 + 2HBr$

(B) $NaCl + H_2SO_4\,(conc.) \rightarrow NaHSO_4 + HCl$

(C) $NaHSO_4 + NaCl \rightarrow Na_2SO_4 + HCl$

(D) $CaF_2 + H_2SO_4\,(conc.) \rightarrow CaSO_4 + 2HF$

Q17. The number of S – S bonds in sulphur trioxide trimer is :–

(A) Three (B) Two

(C) One (D) Zero

Q18. Which of the following statements is unacceptable?

(A) Among $HClO_3$, $HClO_4$, HClO and $HClO_2$, $HClO_3$ is the weakest acid.

(B) Bromide ion can be oxidised to Br_2 by chlorine.

(C) Cl_2O_7 is the anhydride of $HClO_4$.

(D) Among halide ions, F^- has the highest hydration energy.

Q19. Which one of the following does not exist ?

(A) $XeOF_4$ (B) NeF_2

(C) XeF_2 (D) XeF_4

Q20. Which of the following oxyacids of chlorine is least oxidising in nature –

(A) HOCl (B) $HClO_2$

(C) $HClO_3$ (D) $HClO_4$

Q21. The oxidising power in the oxyacids is in the order shown below

$HClO > HClO_2 > HClO_3 > HClO_4$

This can be explained by :-

(A) The stability of anions increasing from ClO^- to ClO_4^-

(B) The oxidising power increasing from ClO^- to ClO_4^-

(C) The acidic character of oxyacids increasing with the increasing in oxidation number

(D) The hydrogen of the oxyacid which is attached to oxygen and which is a part of the – OH group.

Q22. The disproportionated products obtained on reaction of an element (P_4) and NaOH, gives X and Y compounds. X and Y.

(A) PH_3,H_2O (B) PH3, NaH_2PO_3

(C) PH_3, Na_2HPO_2 (D) PH_3, NaH_2PO_2

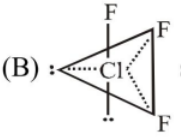
Q23. An element (X) forms compounds of the formula XCl_3, X_2O_5 and Ca_3X_2, but does not form XCl_5. Which of the following is the element –

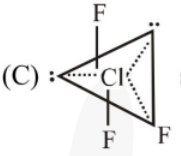
(A) B (B) Al

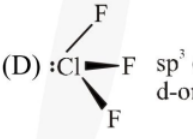
(C) N (D) P

Q24. Indicate for the ClF_3 molecule, the shape and the type of hybridization (of the Cl-atom)

(A)  sp^2

(B)  sp^3d

(C)  sp^3d

(D)  sp^3 one lone pair in d-orbital

Q25. $x\,NH_2OH + y\,NaBrO_3 \rightarrow$ Products are NaBr, H_2O, O_2 and N_2. What are x and y ?

(A) 12, 4 (B) 4, 10

(C) 6, 12 (D) 4, 8

Exercise - 2A — *Brain Booster*

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Which of the following statements is/are correct –

(A) A three electron bond is about half as strong as a covalent bond

(B) Odd electron molecules are usually extremely reactive and tend to dimerise

(C) NO is unusually stable, but it is quite reactive

(D) NO acts as a ligand

Q2. Which of the following statements is/are correct for 15 group elements –

(A) The electropositive character increases within the group

(B) All elements except nitrogen are solids and have allotropic forms

(C) All nitrogen halides, except NF_3 are unstable and often explode

(D) None is correct

Q3. Which of the following statements is/are correct for pentahalides of 15 group elements –

(A) PF_5, PBr_5, AsF_5 and SbF_5 are all known

(B) These molecules have a trigonal bipyramidal shape in gas phase

(C) In the solid state PCl_5 exists as $[PCl_6]^-$ and $[PCl_4]^+$ which have octahedral and tetrahedral structures respectively

(D) PBr_5 splits into $[PBr_4]^+$ and Br^-

Q4. Which of the following statements is/are correct –

(A) Fluorine brings out the maximum valency of six with S, Se and Te

(B) SF_6, SeF_6 and TeF_6 are colourless gases and have octahedral structure

(C) SF_6, SeF_6 and TeF_6 all have low boiling points

(D) None is correct

Q5. Which of the following statements is/are correct –

(A) SCl_2 is prepared by saturating S_2Cl_2 with chlorine.

(B) SCl_2 is angular in shape.

(C) The monohalides S_2F_2, Se_2Cl_2 and Se_2Br_2 are hydrolysed slowly and tend to disproportionate.

(D) None is correct.

Q6. Which of the following statements is/are correct –

(A) SO_2 acts as a mild reducing agent in acidic solution and a strong reducing agent in alkaline solution.

(B) TeO_2 is extremely basic in nature

(C) The basic nature of oxides SO_2, SeO_2 and TeO_2 increases down the group

(D) SeO_2 forms H_2SeO_3 with water, and it can be obtained in crystalline state

Q7. Which of the following statements regarding ozone is/are correct –

(A) $2O_3 \longrightarrow 3O_2$ is endothermic process

(B) Small amount of O_3 is formed in the electrolysis of dilute H_2SO_4

(C) O–O bonds in ozone have considerable double bond character

(D) None is correct

Q8. Which of the following statements regarding halogens is/are correct –

(A) Halogens act as oxidising agents

(B) F is the strongest oxidising agent

(C) The reaction of fluorine with water is strongly endothermic

(D) The oxidation of water by chlorine is thermodynamically possible, but his reaction does not occur

Q9. Which of the following option is/are correct?

(A) vapour phase of XeF_6 is covalent and hybridisation is sp^3d^3

(B) Solid phase of XeF_6 is ionic and hybridisation is sp^3d^2

(C) XeF_2 is linear in shape

(D) XeO_2F_2 follows bent rule and structure is see-saw.

Q10. Pick out the correct statement for XeF_6

(A) XeF_6 is hydrolysed partially to form $XeOF_4$

(B) It react with SiO_2 to form $XeOF_4$

(C) On complete hydrolysis, it forms XeO_3

(D) It acts as F^- acceptor when treated with alkali metal fluoride, but cannot act as F^- donor to form complexes.

Q11. Which of the following statements is/are correct –

(A) No polysulphuric acid higher than pyrosulphuric acid is known.

(B) No peroxo acids of Se and Te are known.

(C) Peroxodisulphuric acid, $H_2S_2O_8$ is a powerful oxidising agent.

(D) Hydrolysis of $H_2S_2O_8$ gives Caro's acid.

Q12. Which of the following statements are true:-

(A) Perchlorates are strong oxidising agents and frequently used in making fireworks and flares.

(B) Mixtures of NH_4ClO_4 and $KClO_4$ with carbon or various organic compounds are used as explosives and rocket fuels.

(C) -ic acids are stronger than -ous acids.

(D) None of these.

Q13. Hydrolysis process not occures in

(A) CCl_4 (B) NF_3

(C) PCl_5 (D) $AsCl_3$

Q14. Which of the following statements is/are correct –

(A) $TeCl_4$ has trigonal bipyramidal geometry with one position occupied by a lone pair.

(B) $TeCl_4$ reacts with HCl to form $[TeCl_6]^{2-}$ complex ion.

(C) $[TeCl_6]^{2-}$ is isomorphous with $[SiF_6]^{2-}$ and $[SbCl_6]^-$.

(D) None is correct.

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph For. Q.15 to 17

There are some deposits of nitrates and phosphates in earth's crust. Nitrates are more soluble in water. Nitrates are difficult to reduce under the laboratory conditions but microbes do it easily. Ammonia forms large number of complexes with transition metal ions. Hybridization eaily explains the ease of sigma donation capability of NH_3 and PH_3. Phosphine is a flammable gas and is prepared from white phosphorus.

Q15. Among the following, the correct statement is –

(A) Phosphates have no biological significance in humans.

(B) Between nitrates and phosphates, phosphates are less abundant in earth's crust.

(C) Between nitrates and phosphates, nitrates are less abundant in earth's crust.

(D) Oxidation of nitrates is possible in soil.

Q16. Among the following, the correct statement is –

(A) Between NH_3 and PH_3, NH_3 is a better electron donor because the lone pair of electrons occupies spherical 's' orbital and is less directional.

(B) Between NH_3 and PH_3, PH_3 is a better electron donor because the lone pair of electrons occupies sp^3 orbital and is more directional.

(C) Between NH_3 and PH_3, NH_3 is a better electron donor because the lone pair of electrons occupies sp^3 orbital and is more directional.

(D) Between NH_3 and PH_3, PH_3 is a better electron donor because the lone pair of electrons occupies spherical 's' orbital and is less directional.

Q17. White phosphorus on reaction with NaOH gives PH_3 as one of the products. This is a –

(A) dimerization reaction

(B) disproportionation reaction

(C) condensation reaction

(D) precipitation reaction

Paragraph For. Q.18 to 21

$$\underset{\text{(organic compound)}}{A} + O_2 \longrightarrow X + Y + Z$$

Compound (A) in pure form does not give ppt. with $AgNO_3$ solution. A mixture containing 70% of (A) and 30% of ether is used as an anaesthetic. Compound (X) and (Y) are oxides while (Z) is a pungent smelling gas. (X) is a neutral oxide which turns cobalt chloride paper pink. Compound (Y) turns lime water milky and produces an acidic solution with water.

Q18. Compound (A) is –

(A) $CHCl_3$ (B) H_2O

(C) CO_2 (D) Cl_2

Q19. Compound (X) is –

(A) $CHCl_3$ (B) Cl_2

(C) CO_2 (D) H_2O

Q20. Compound (Y) is –

(A) $CHCl_3$ (B) H_2O

(C) CO_2 (D) Cl_2

Q21. Compound (Z) is –

(A) $CHCl_3$ (B) H_2O

(C) CO_2 (D) Cl_2

Paragraph For. Q.22 to 24

The noble gases have closed-shell electronic configuration and are monoatomic gases under normal conditions. The low boiling points of the lighter noble gases are due to weak dispersion forces between the atoms and the absence of other interatomic interactions.

The direct reaction of xenon with fluorine leads to a series of compounds with oxidation numbers +2, +4 and +6. XeF_4 reacts violently with water to give XeO_3. The compounds of xenon exhibit rich stereochemistry and their geometries can be deduced considering the total number of electron pairs in the valence shell.

Q22. Argon is used in arc welding because of its-

(A) low reactivity with metal

(B) ability to lower the melting point of metal

(C) flammability

(D) high calorific value

Q23. The structure of XeO_3 is -

(A) linear

(B) planar

(C) pyramidal

(D) T-shaped

Q24. XeF_4 and XeF_6 are expected to be -

(A) oxidizing

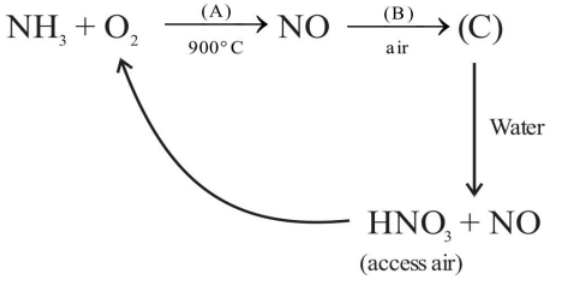
(B) reducing

(C) unreactive

(D) strongly basic

Paragraph For. Q.25 to 27

The following flow diagram represents the industrial preparation of nitric acid from ammonia:



Answer the questions given below:

Q25. Which line of entry describes the undefined reagents, products and reaction conditions?

	A	B	C
(A)	catalyst	R.T. (25°C)	NO_2
(B)	catalyst	R.T. (25°C)	N_2O
(C)	catalyst	high pressure	NO_2
(D)	high pressure	catalyst	N_2O_3

Q26. Formation of HNO_3 when (C) is dissolved in H_2O takes place through various reactions. Select the reaction not observed in this step.

(A) $NO_2 + H_2O \longrightarrow HNO_3 + HNO_2$

(B) $HNO_2 \longrightarrow H_2O + NO + NO_2$

(C) $NO_2 + H_2O \longrightarrow HNO_3 + NO$

(D) none of these

Q27. Nitrogen dioxide is dissolved in water to produce

(A) HNO_3 and HNO_2 (B) only HNO_3

(C) only HNO_2 (D) HNO_2 and N_2

Paragraph For. Q.28 to 32

A sulphate of a metal (A) on heating evolves two gases (B) and (C) and an oxide (D). Gas (B) turns $K_2Cr_2O_7$ paper green while gas (C) forms a trimer in which there is no S – S bond. Compound (D) with HCl, forms a Lewis acid (E) which exists as a dimer.

Q28. Compound (A) is -

(A) $FeSO_4$ (B) SO_2

(C) SO_3 (D) Fe_2O_3

Q29. Compound (B) is -

(A) $FeSO_4$ (B) SO_2

(C) SO_3 (D) Fe_2O_3

Q30. Compound (C) is -

(A) $FeSO_4$ (B) SO_2

(C) SO_3 (D) Fe_2O_3

Q31. Compound (D) is -

(A) $FeSO_4$ (B) SO_2

(C) SO_3 (D) Fe_2O_3

Q32. Compound (E) is -

(A) $FeCl_3$ (B) SO_2

(C) SO_3 (D) Fe_2O_3

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q33. Match the column correctly –

Column-I	**Column-II**
(A) Best reductant	(P) Fluorine
(B) Arracks glass	(Q) F^-
(C) Highest. hydrated halide negative ion	(R) HF
(D) Tri-iodide ion	(S) HI
(E) Does not show +1 oxidation state	(T) Linear species

Q34. Match the column –

Column-I (Chemical reaction)	**Column-II (Name of process)**
(A) $4NH_3 + 5O_2 \xrightarrow{800°C/Pt} 4NO + 6H_2O$	(P) Contact process
(B) $4HCl + O_2 \xrightarrow[450-500°C/V_2O_5]{3230°C/CuCl_2} 2Cl_2 + 2H_2O$	(Q) Ostwald's process
(C) $2SO_2 + O_2 \longrightarrow 2SO_3$	(R) Deacon's process
(D) $2N_2 + 3H_2 \xrightarrow{Fe+Mo} 2NH_3$	(S) Haber's process

Q35. Match the column –

Column-I	**Column II**
(A) N_2O_4	(P) Produces mixed acids on reaction with water as final product
(B) Cl_2O_6	(Q) Exists as ion pair in the solid state
(C) SO_2Cl_2	(R) Paramagnetic in monomeric form
	(S) Stronger dehydrating agent compared to conc. H_2SO_4

ASSERTION AND REASON

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

Q36. **Statement 1 :** Elemental phosphorus exist as P_2 under ordinary conditions.

Statement 2 : Phosphorus does not form $p_\pi - p_\pi$ multiple bonds.

Q37. **Statement 1 :** Chlorine gas disporoportionates in hot and conc. NaOH solution.

Statement 2 : NaCl and NaOCl are formed in the above reaction.

Answer Key

EX-1									
1	2	3	4	5	6	7	8	9	10
B	C	B	B	C	C	C	A	D	B
11	12	13	14	15	16	17	18	19	20
A	A	C	A	D	D	B	B	C	A
21	22	23	24	25	26				
A	B	C	C	C	B				

Ex-1A

1. B,D **2.** A,B **3.** A,B,C,D **4.** A,B,C **5.** B,C,D

6. A,B,C,D **7.** A **8.** B **9.** C **10.** D

11. A → Q ; B → R ; C → S ; D → P

12. A→ S ; B→ Q ; C→ P ; D→ R

13. D **14.** A **15.** A **16.** A **17.** A

18. (a) $I_2 < F_2 < Br_2 < Cl_2$

(b) HF < HCl < HBr < HI

(c) $BiH_3 < SbH_3 < AH_3 < PH_3 < NH_3$

19. (i) $N_2O_5 > P_2O_5 > AS_2O_5 > Sb_2O_5 > Bi_2O_5$

(ii) $PI_3 < PBr_3 < PCl_3 < PF_3$

(iii) $H_3PO_4 < H_3PO_3 < H_3PO_2$

(iv) $NI_3 > NBr_3 > NCl_3 > NF_3$

(v) $SbH_3 < ASH_3b< PH_3(i< NH_3v)$

EX-2									
1	2	3	4	5	6	7	8	9	10
B	D	D	D	C	C	D	C	B	A
11	12	13	14	15	16	17	18	19	20
A	B	D	D	B	A	D	A	B	D
21	22	23	24	25					
A	D	C	C	A					

Ex-2A

1. A,B,C,D **2.** A,B,C **3.** A,B,C,D **4.** A,B,C **5.** A,B,C

6. A,C,D **7.** A,B,C **8.** A,B **9.** A,B,C,D **10.** A,B,C

11. A,B,C **12.** A,B,C **13.** A,B,C **14.** A,B,C **15.** C

16. C **17.** B **18.** A **19.** D **20.** C

21. D **22.** A **23.** C **24.** A **25.** A

26. D **27.** A **28.** A **29.** B **30.** C

31. D **32.** A

33. A→ S, B→ R, C→ Q, D→ T, E→ P

34. A→ Q, B→ R, C→ P, D→S

35. A→ PQR, B→ PQR, C→P

36. D **37.** C

-: Important Notes & Formulas :-